

Heart Disease Prediction based on AI

Human-AI Cooperation: Human in the Loop, Explainability, Trustworthiness and Transparency in AI

Benjamim Moreira

José Filipe

José Cunha

Universidade de Coimbra

uc2020261856@student.uc.pt, josepedrofilipe1@gmail.com, josecunha.8989@gmail.com

Abstract

Cardiovascular diseases are the leading cause of death globally, accounting for 31% of all deaths according to the World Health Organization[3]. Early detection and intervention are crucial in reducing mortality rates and improving patient outcomes. This project consists on developing a machine learning model for heart disease prediction, utilizing medical data to identify patients at risk. By leveraging Human-Centered AI (HCAI) principles, the system will integrate Human-in-the-Loop (HITL) approaches to ensure that medical professionals can interact with and provide feedback on the model's predictions.

By focusing on both predictive accuracy and human collaboration, this project aims to create a decision support tool that improves the reliability and transparency of AI in medical diagnostics.

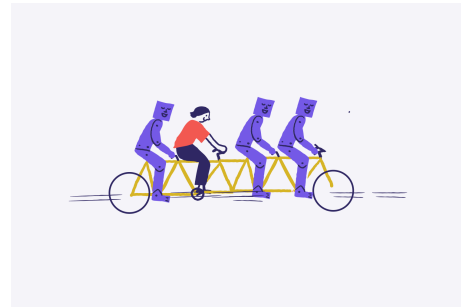


Figure 1: Human AI Collaboration

1 Introduction

Heart disease is a leading cause of mortality worldwide, affecting millions of individuals each year. Despite advances in medical research and diagnostics, heart disease often remains asymptomatic until it reaches advanced stages, making early detection a crucial factor in reducing the global burden of cardiovascular conditions. The ability to accurately predict the likelihood of heart disease in individuals at an early stage can lead to timely medical interventions, potentially saving lives and reducing healthcare costs.

Recent advances in machine learning have provided promising tools for predicting heart disease by analyzing large datasets of medical information.[2] These models have the potential to uncover patterns that are difficult to detect through traditional diagnostic methods, offering more accurate risk assessments. However, the widespread adoption of AI in healthcare requires that these systems not only deliver accurate predictions but also align with human values, ensuring trust, transparency, offer a clear explanation of the process carried out and cooperation between machines and healthcare professionals.

This project seeks to develop a heart disease prediction model that integrates principles of Human-Centered AI (HCAI), with a focus on fostering human-AI cooperation. The model will utilize machine learning techniques to analyze patient data and predict heart disease risk, while incorporating Human-in-the-Loop (HITL) methods to allow medical professionals to interact with and refine the system's predictions.

2 Approach

The current approach in the base project for heart disease prediction likely involves traditional machine learning models, such as logistic regression, decision trees, or even neural networks, applied to clinical datasets containing features like age, blood pressure, cholesterol levels, and more. This conventional method focuses on building a predictive model, training it with labeled data, and evaluating performance using metrics such as accuracy, precision, and recall.

However, one significant challenge with these models, especially more complex ones like neural networks or ensemble methods (e.g., random forests, XGBoost), is interpretability. These models often function as "black boxes," making it difficult to understand how decisions are made. This lack of transparency is particularly problematic in healthcare contexts, where it's critical for physicians and healthcare professionals to comprehend the rationale behind a model's predictions.

2.1 Applying SHAP

In our project, instead of following the original approach, we will apply the SHAP (SHapley Additive exPlanations) method to improve the interpretability of our model's results. SHAP is an advanced explanation technique that provides clear insights into the contribution of each feature to the

model's predictions. Based on Shapley values from cooperative game theory, SHAP assigns a specific importance score to each feature, explaining how much each factor contributes to a given prediction.[1].

By incorporating SHAP into the project, we will maintain the predictive power of our model, but crucially, we will also be able to understand and quantify how individual factors (e.g., age, blood pressure, cholesterol) influence the prediction of heart disease risk. This adds a layer of transparency to the model, providing healthcare professionals with a clear rationale for how patient-specific features impact the predicted outcome.

This approach not only enhances the interpretability of the predictions but also fosters greater trust and reliability in the model, making it more suitable for real-world clinical applications. It aligns with the principles of Human-Centered AI (HCAI) by ensuring that the AI system supports human decision-making in a clear and explainable manner, facilitating collaboration between AI and medical experts.

3 Materials

The project will utilize the Z-Alizadeh Sani dataset, sourced from the UCI Machine Learning Repository, which contains 303 medical records of patients who presented with chest pain. Each record comprises 55 features, including demographic information, physical examinations, ECG results, laboratory tests, and echocardiography.[4] Although the Z-Alizadeh Sani dataset includes only 303 records, which may seem limited for large-scale machine learning applications, the quality and diversity of the features make it a strong basis for prediction. Initial evaluations using techniques such as cross-validation have demonstrated promising results, with high accuracy, recall, and precision metrics. Given these outcomes, the dataset is deemed sufficient for the current project, and no further data collection is necessary to achieve the project's objectives.

To implement the machine learning models and ensure the interpretability of the predictions, at least for now, several frameworks and tools will be employed:

- **Python:** The main programming language for this project, due to its extensive libraries for data analysis, machine learning, and visualization.
- **SHAP (SHapley Additive exPlanations):** SHAP will be integrated to explain the output of the models, providing interpretability to the black-box predictions by assigning importance values to features for each prediction.
- **Pandas and NumPy:** These libraries will be used for data manipulation and numerical operations, enabling efficient processing of the dataset and feature engineering.
- **Matplotlib and Seaborn:** Visualization libraries that will be used to create plots and graphs for data analy-

sis and model performance evaluation, helping in understanding the distributions of features and the impact of predictions.

4 Evaluation

The heart disease prediction model will produce binary outcomes, classifying patients into two categories: "No Heart Disease" or "Potential for Heart Disease". In cases classified as "Potential for Heart Disease," the prediction will trigger a human verification process, allowing healthcare professionals to review the AI's findings and confirm or adjust the diagnosis, ensuring a reliable and trustworthy system. These metrics will provide a robust evaluation of the model's ability to correctly identify individuals at risk, ensuring a balance between minimizing false positives (over-diagnosis) and false negatives (missed diagnoses).

Additionally, graphical tools will be employed to better understand the weight and impact of each feature on the model's predictions. These visualizations, such as feature importance plots and SHAP value graphs, will help to illustrate how individual factors (e.g., age, cholesterol, blood pressure) contribute to the final prediction. This approach will enhance the transparency of the model and assist healthcare professionals in interpreting the AI's decision-making process, further increasing trust in the system.

References

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